



SQL OF THE DAY

mind the bonus

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StrataScratch · #10077

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THE PROBLEM

Find the average total compensation (salary + bonus) by title and gender. Only include employees who received at least one bonus, and one employee can receive several.

averages mislead...

Medium · PostgreSQL





THE SOLUTION

```
WITH cte_total_bonus AS (  
    SELECT worker_ref_id,  
           SUM(bonus) AS total_bonus  
    FROM sf_bonus  
    GROUP BY worker_ref_id  
)  
SELECT e.employee_title,  
       e.sex,  
       AVG(e.salary + b.total_bonus) AS avg_comp  
FROM sf_employee e  
INNER JOIN cte_total_bonus b  
    ON e.id = b.worker_ref_id  
GROUP BY 1, 2;
```





HOW IT WORKS

AGGREGATE FIRST

Sum every bonus per employee with `SUM(bonus) GROUP BY worker_ref_id`, so multiple bonus rows do not multiply the salary.

INNER JOIN FILTERS

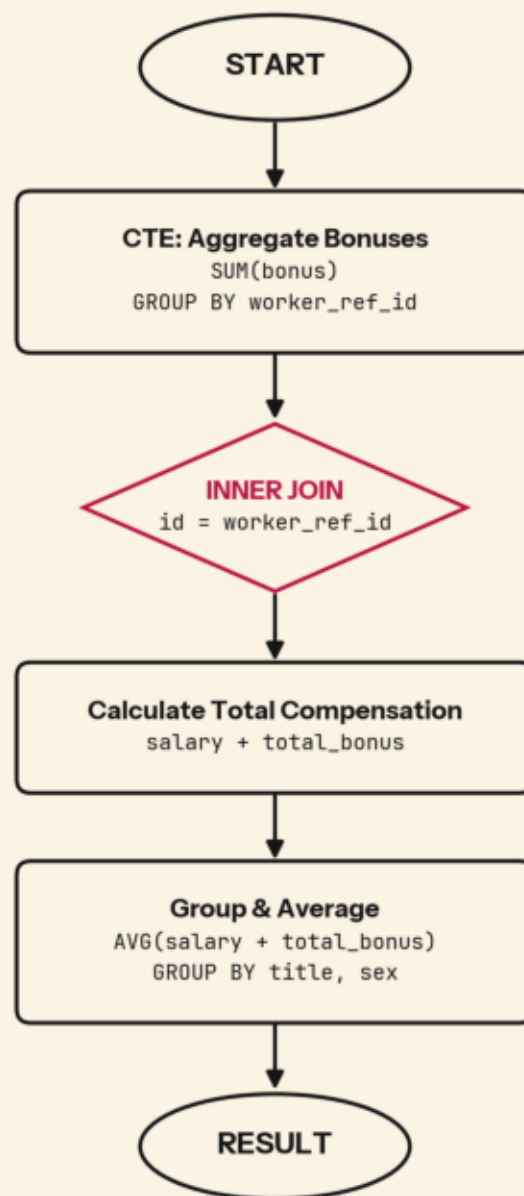
The join to the bonus CTE drops anyone who never received a bonus, no `WHERE` clause needed.

GROUP AND AVERAGE

Average salary + total_bonus, grouped by title and sex.



THE LOGIC





WHY IT MATTERS

01

Pay-equity exposure

Average comp by title and gender is the first screen for a pay gap that carries legal and retention risk if ignored.

02

The average can hide it

Two roles can share a mean while one masks a wide spread, so an equity call on the average alone can clear a gap that is really there.

03

Benchmarking

Role averages against market data flag where pay sits below market and retention risk is building.





KEY TAKEAWAYS

- 1 Aggregate child rows in a CTE before joining, or the join multiplies the parent value.
- 2 INNER JOIN on the bonus table filters out non-bonus employees on its own.
- 3 Averages hide spread. For equity work, read the quartiles with `percentile_cont()`.

follow for more SQL problems

questions? nivanrs@gmail.com

